Lab 1.3

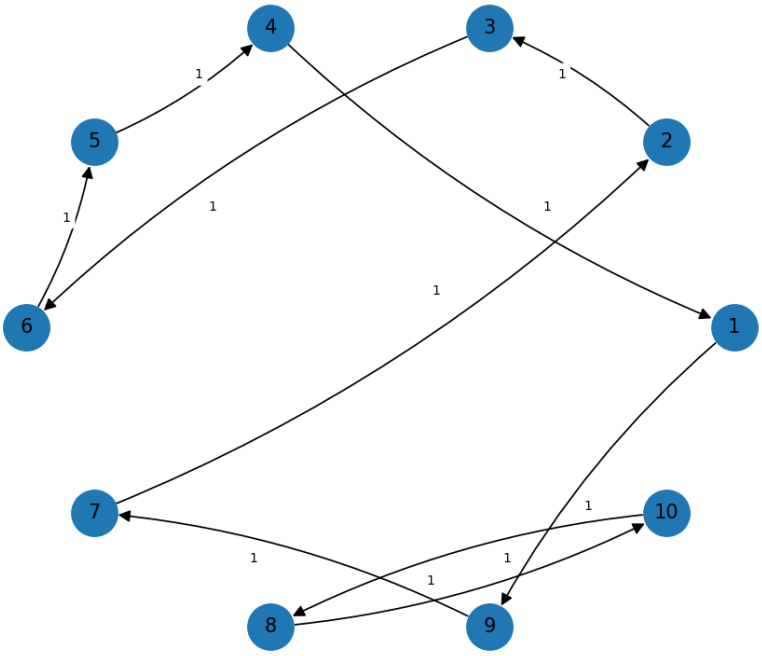
Adomas Bieliūnas
2413914 22 var

2026

1.2

$$M = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$x_k = Mx_{k-1} = MMx_{k-2} = M^k x_0$$

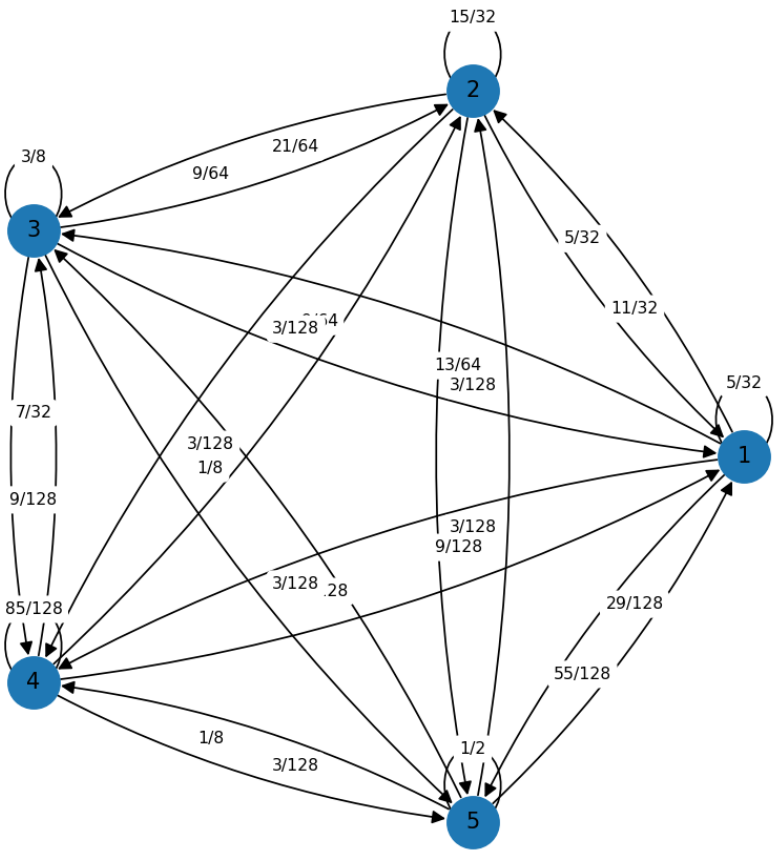


2

$$N = 5, k = -2$$

$$\text{Atsitiktinė matrica } M = \begin{bmatrix} \frac{5}{32} & \frac{11}{32} & \frac{13}{64} & \frac{9}{128} & \frac{29}{128} \\ \frac{5}{32} & \frac{15}{32} & \frac{21}{64} & \frac{3}{128} & \frac{3}{128} \\ \frac{9}{64} & \frac{9}{64} & \frac{3}{8} & \frac{7}{32} & \frac{1}{8} \\ \frac{15}{128} & \frac{3}{128} & \frac{9}{128} & \frac{85}{128} & \frac{1}{8} \\ \frac{55}{128} & \frac{3}{128} & \frac{3}{128} & \frac{3}{128} & \frac{1}{2} \end{bmatrix}$$

$$\text{Būsena po k žingsnių } M^k = \begin{bmatrix} \frac{19815468005}{231678841} & -\frac{11054355483}{231678841} & -\frac{123879823}{22420533} & -\frac{2254274129}{695036523} & -\frac{6497917467}{231678841} \\ -\frac{12019659857}{695036523} & \frac{13197134767}{695036523} & -\frac{1017150637}{67261599} & \frac{13803143437}{2085109569} & \frac{1809023461}{231678841} \\ -\frac{3222802513}{695036523} & -\frac{2222878801}{695036523} & \frac{1134624595}{67261599} & -\frac{13964078323}{2085109569} & -\frac{309681179}{231678841} \\ -\frac{600989723}{231678841} & \frac{529342437}{231678841} & -\frac{14929029}{7473511} & \frac{758635493}{231678841} & \frac{7490533}{231678841} \\ -\frac{13901978651}{231678841} & \frac{7098606565}{231678841} & \frac{151929457}{22420533} & \frac{727049135}{695036523} & \frac{5222763493}{231678841} \end{bmatrix}$$



3

$$N = 4, k = 2, i = 2, j = 4, T = 4$$

$$\text{Atsitiktinė matrica } M =$$

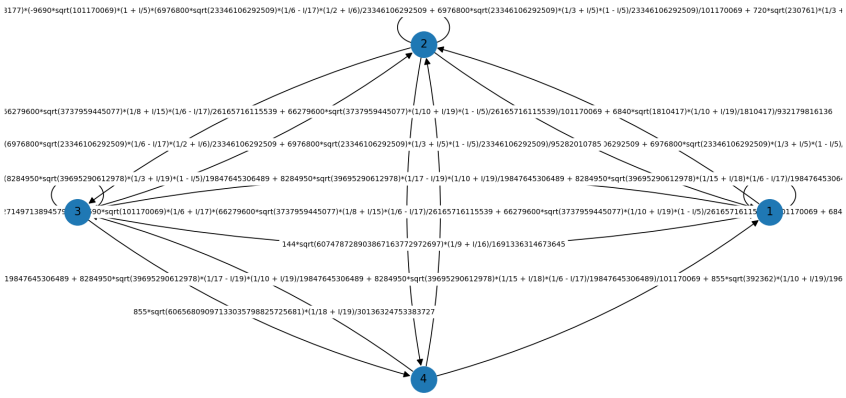
$$\begin{bmatrix} \frac{95\sqrt{101170069}(17+6i)}{101170069} & 0 & 0 & 0 \\ \frac{1938\sqrt{101170069}(5+i)}{101170069} & 0 & 0 & 0 \\ \frac{30\sqrt{101170069}(19+17i)}{101170069} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} +$$

$$\begin{bmatrix} 0 & \frac{\sqrt{607478728903867163772972697}(487522944+135406656i)}{1316250858905679824088 5} & 0 & 0 \\ 0 & \frac{\sqrt{607478728903867163772972697}(-16844016-1976112i)}{26325017178113596481 77} & 0 & 0 \\ 0 & \frac{\sqrt{607478728903867163772972697}(9171072-15250464i)(2632501717811359648177-5265003435622719296354i)}{3465032647139879711759620659253952609711664 5} & 0 & 0 \\ 0 & \frac{\sqrt{607478728903867163772972697}(16+9i)}{1691336314673645} & 0 & 0 \end{bmatrix} +$$

$$\begin{bmatrix} 0 & 0 & \frac{\sqrt{606568090971330357988257256 81}(705716145+356995161i)}{2345303119004799726205 51} & 0 \\ 0 & 0 & \frac{135\sqrt{606568090971330357988257256 81}(-1236861-2469602i)(1-i)}{9381212476019198904822 04} & 0 \\ 0 & 0 & \frac{855\sqrt{606568090971330357988257256 81}(-302659-756162i)(5-i)}{1219557621882495857626865 2} & 0 \\ 0 & 0 & \frac{\sqrt{606568090971330357988257256 81}(95+90i)}{60272649506767454} & 0 \end{bmatrix} +$$

$$\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{1170016465522803709314805654353}(-68679186+192608966i)(458766478806190244768001-305844319204126829845334i)}{3040074296656664265993080702778433032006751075 57} \\ 0 & 0 & 0 & \frac{\sqrt{1170016465522803709314805654353}(-74576620-153276840i)(1376299436418570734304003-917532957612380489536002i)}{2736066866990997839393772632500589728806075968013} \\ 0 & 0 & 0 & \frac{\sqrt{1170016465522803709314805654353}(234967262+1370359553i)(917532957612380489536002-1376299436418570734304003i)}{2736066866990997839393772632500589728806075968013} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Būsena po k žingsnių, G(A) pertvarko stulpelius pagal 1 stulpelį su GS procesu $(G(M))^k$



Tikimybė pereiti nuo būsenos i į būseną j per T: 0.0097

4.9

$$|\psi\rangle = \begin{bmatrix} 7+5i \\ -1-3i \\ 6i \\ -5+2i \\ 4-7i \\ 6-8i \\ -1+7i \\ -7-8i \end{bmatrix}$$

$$P(z) = |z|^2$$

$$P = \begin{bmatrix} |7+5i|^2 \\ |-1-3i|^2 \\ |6i|^2 \\ |-5+2i|^2 \\ |4-7i|^2 \\ |6-8i|^2 \\ |-1+7i|^2 \\ |-7-8i|^2 \end{bmatrix} = \begin{bmatrix} 74 \\ 10 \\ 36 \\ 29 \\ 65 \\ 100 \\ 50 \\ 113 \end{bmatrix}$$

5.9

$$|\psi\rangle = \begin{bmatrix} 7+5i \\ -1-3i \\ 6i \\ -5+2i \\ 4-7i \\ 6-8i \\ -1+7i \\ -7-8i \end{bmatrix}, \ |\phi\rangle = \begin{bmatrix} 7+3i \\ -6-7i \\ 5+5i \\ -4-i \\ -6-3i \\ 3-6i \\ -6+5i \\ -8-2i \end{bmatrix}$$

$$|\hat{\psi}\rangle = \frac{|\psi\rangle}{|||\psi\rangle||}, |\hat{\phi}\rangle = \frac{|\phi\rangle}{|||\phi\rangle||}$$

$$\text{Tranzicijos amplitudė} = \langle\hat{\phi}|\hat{\psi}\rangle = \sum_{d=1}^{n} \hat{\phi}_d \overline{\hat{\psi}_d}$$

$$|||\psi\rangle|| = \sqrt{\sum |\psi_i|^2} = \sqrt{\sum P_i}$$

$$|||\psi\rangle|| = \sqrt{74+10+36+29+65+100+50+113} = \sqrt{477}$$

$$|||\phi\rangle|| = \sqrt{58+85+50+17+45+45+61+68} = \sqrt{429}$$

$$|\hat{\psi}\rangle = \frac{1}{\sqrt{477}} \begin{bmatrix} 7+5i \\ -1-3i \\ 6i \\ -5+2i \\ 4-7i \\ 6-8i \\ -1+7i \\ -7-8i \end{bmatrix}$$

$$|\hat{\phi}\rangle = \frac{1}{\sqrt{429}} \begin{bmatrix} 7+3i \\ -6-7i \\ 5+5i \\ -4-i \\ -6-3i \\ 3-6i \\ -6+5i \\ -8-2i \end{bmatrix}$$

$$\langle\hat{\phi}|\hat{\psi}\rangle = \frac{35}{22737} + i\frac{11}{18603}$$

6

$$A = \begin{bmatrix} \xi & 7 & \tau \\ a & b & 9 \\ 0 & \tau & -2 \end{bmatrix}, \; B = \begin{bmatrix} \gamma & 37 & d \\ 1 & d & 0 \\ \xi & c & a \end{bmatrix}$$

$$[A, B] = AB - BA =$$

$$\begin{bmatrix} -37a + \tau\xi + 7 & -37b + c\tau - d\tau + 7d - 7\gamma + 37\xi & a\tau + d\xi + 2d - \gamma\tau - 333 \\ -ad + a\gamma + b + 8\xi & 37a + 9c - 7 & ad + 9a - 9d - \tau \\ -ac + \tau - \xi^2 - 2\xi & -a\tau - bc - 2c + d\tau - 7\xi & -9c - \tau\xi \end{bmatrix}$$